

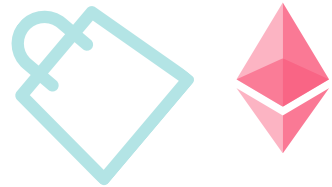


Exploring Customer Engagement & Purchasing Patterns in E-Commerce

DATA 200- Applied Statistical Analysis



Problem statement



Customer engagement plays a crucial role in influencing purchasing behavior, yet it remains unclear which specific engagement actions truly drive customers to complete a purchase. This study aims to identify the key differences between buyers and browsers and determine the most predictive customer actions that lead to purchase conversion.





Dataset

E-commerce Customer Behavior Dataset

Exploring Customer Engagement and Purchasing Patterns in an E-commerce

Data Card

Code (44)

Discussion (2)

Suggestions (0)

About Dataset

Dataset Description: E-commerce Customer Behavior

Overview:

This dataset provides a comprehensive view of customer behavior within an e-commerce platform. Each entry in the dataset corresponds to a unique customer, offering a detailed breakdown of their interactions and transactions. The information is crafted to facilitate a nuanced analysis of customer preferences, engagement patterns, and satisfaction levels, aiding businesses in making data-driven decisions to enhance the customer experience.



Source: <https://www.kaggle.com/datasets/uom190346a/e-commerce-customer-behavior-dataset?resource=download>

Data Exploration

Dataset Overview

- **Total Customers:**
350
- **Total Features:** 11
- **Dataset Type:** Structured tabular dataset
- **Contains both numerical and categorical variables**
- **Focus:** Customer demographics, purchasing behavior, and satisfaction metrics



Features in Dataset



Demographic Features:

Gender

Age

City



Purchase & Spending Behavior

Total spend

Items Purchased

Discount Applied

Days Since Last purchase



Membership Information:

Membership type



Customer Feedback:

Average Rating

Satisfaction Level



Model Selection

Model selection was based on the type of outcome variable and research objective. Since Purchase (Yes/No) is binary, Logistic Regression was selected as the main predictive model. ANOVA and Multiple Linear Regression were used to support comparison and relationship analysis.



Why These Models?

Chosen based on variable type and study objective



Logistic Regression

Predicts probability of purchase (binary outcome)



ANOVA

Compares engagement levels between buyers and browsers



Multiple Linear Regression

Analyzes relationships among engagement variables





Feature Engineering



Feature engineering focused on selecting meaningful predictors that strongly influence purchasing behavior. Variables were chosen based on exploratory data analysis and business relevance to improve model performance and interpretability.

Cart_Additions

Direct indicator of
purchase intent

Pages_Visited & Items_Viewed

Measures of
engagement level

Previous_Purchases

Indicator of customer
loyalty



Selected key features to enhance predictive accuracy



Strongest predictor:

Cart_Additions, indicating direct purchase intent

Model Analysis & Evaluation



Model demonstrated

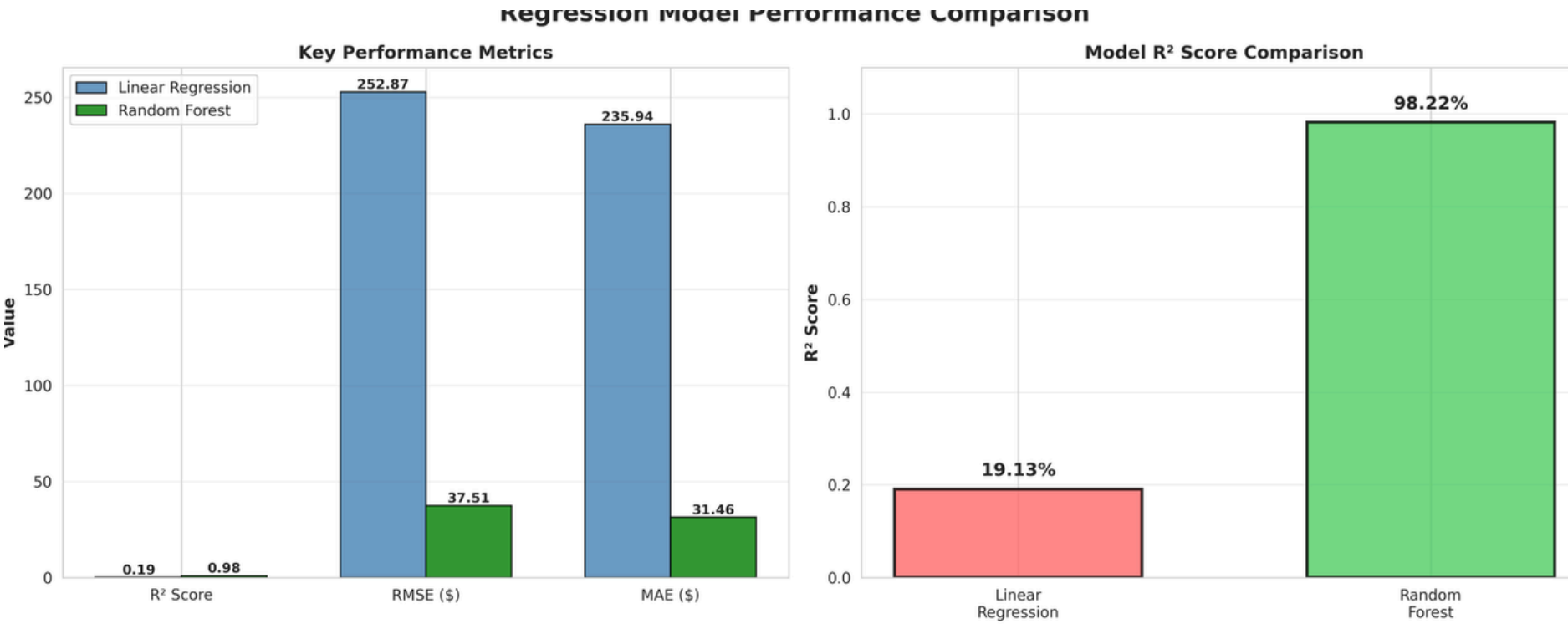
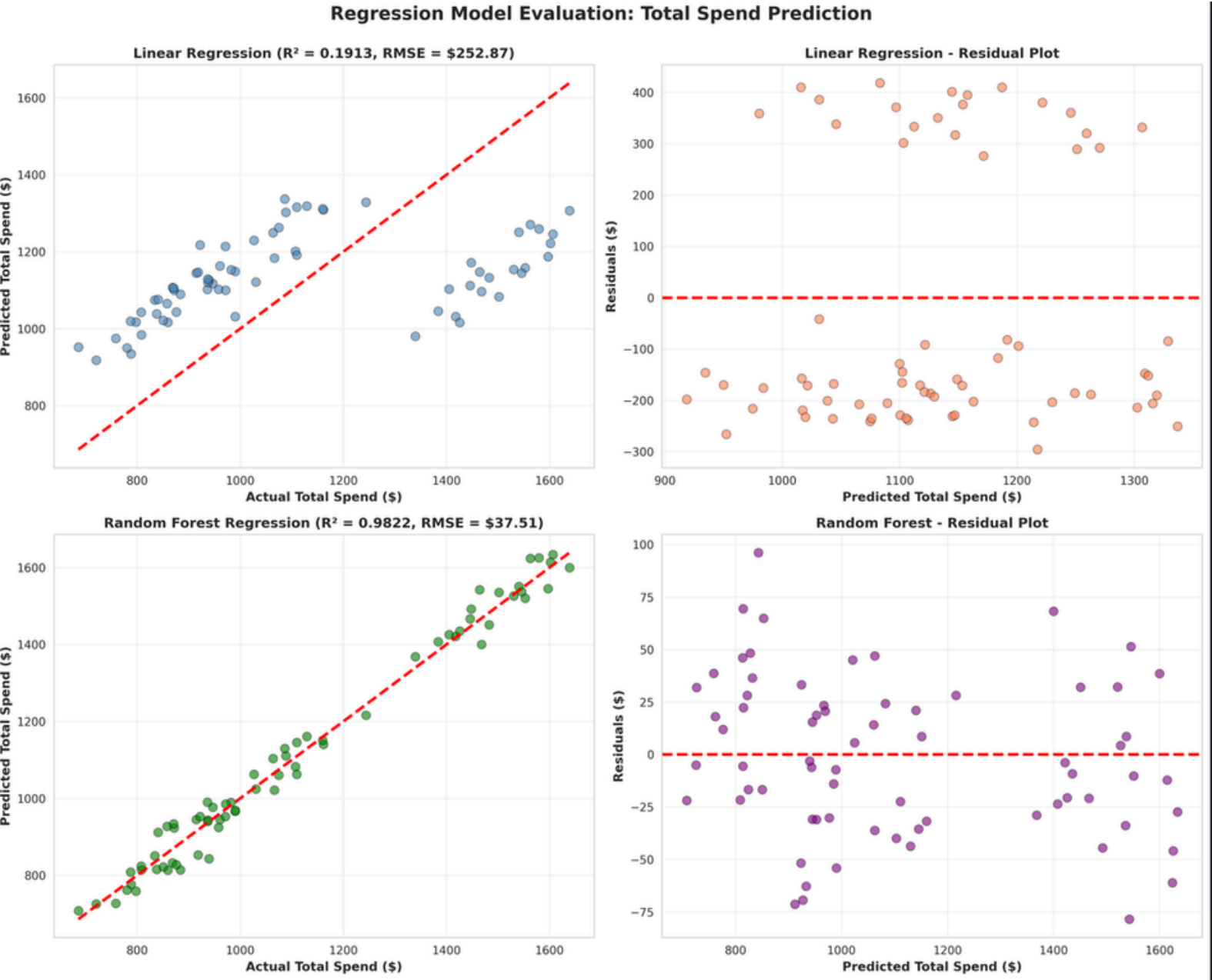
It demonstrates acceptable explanatory power (R^2 shows meaningful variability explained)

Diagnostic tests

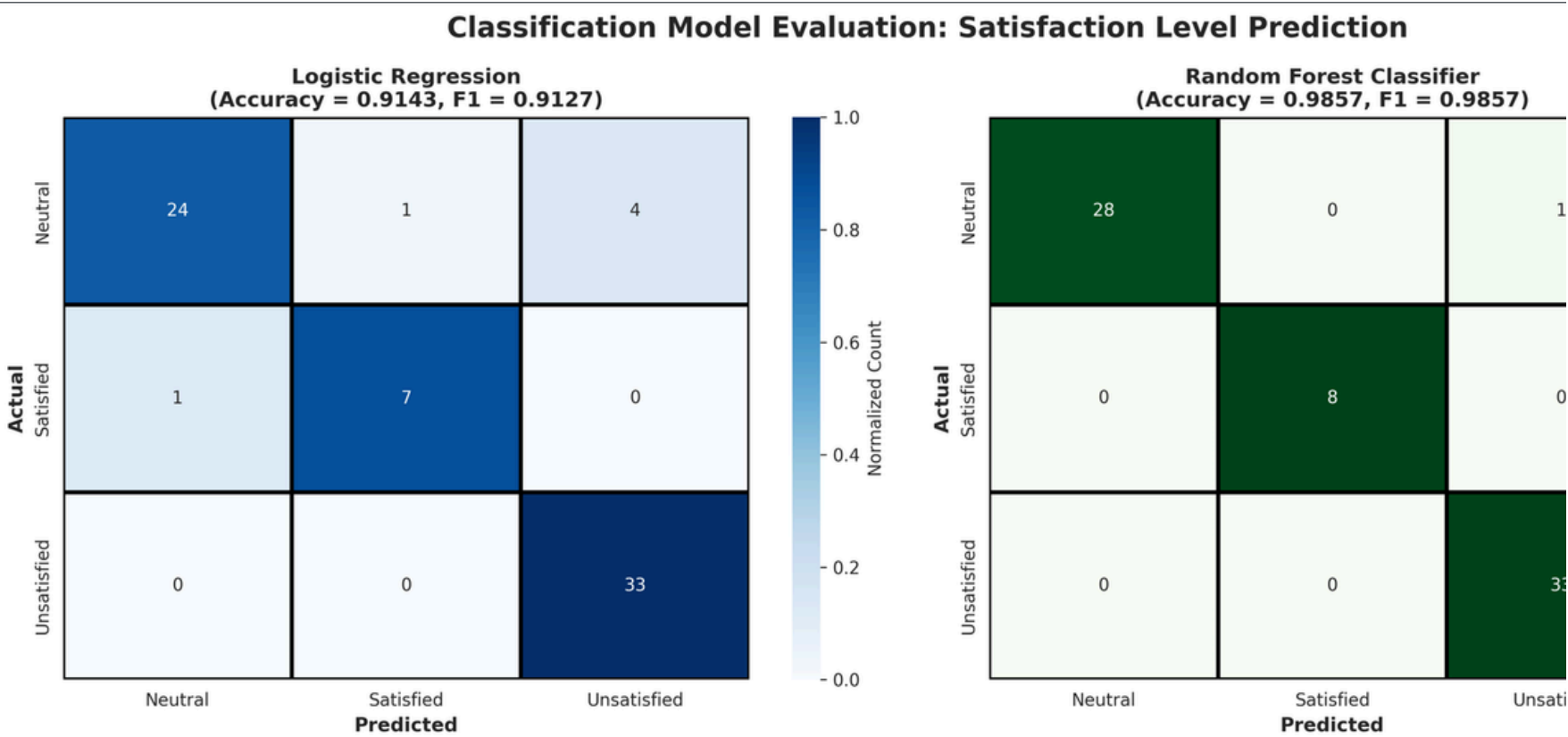
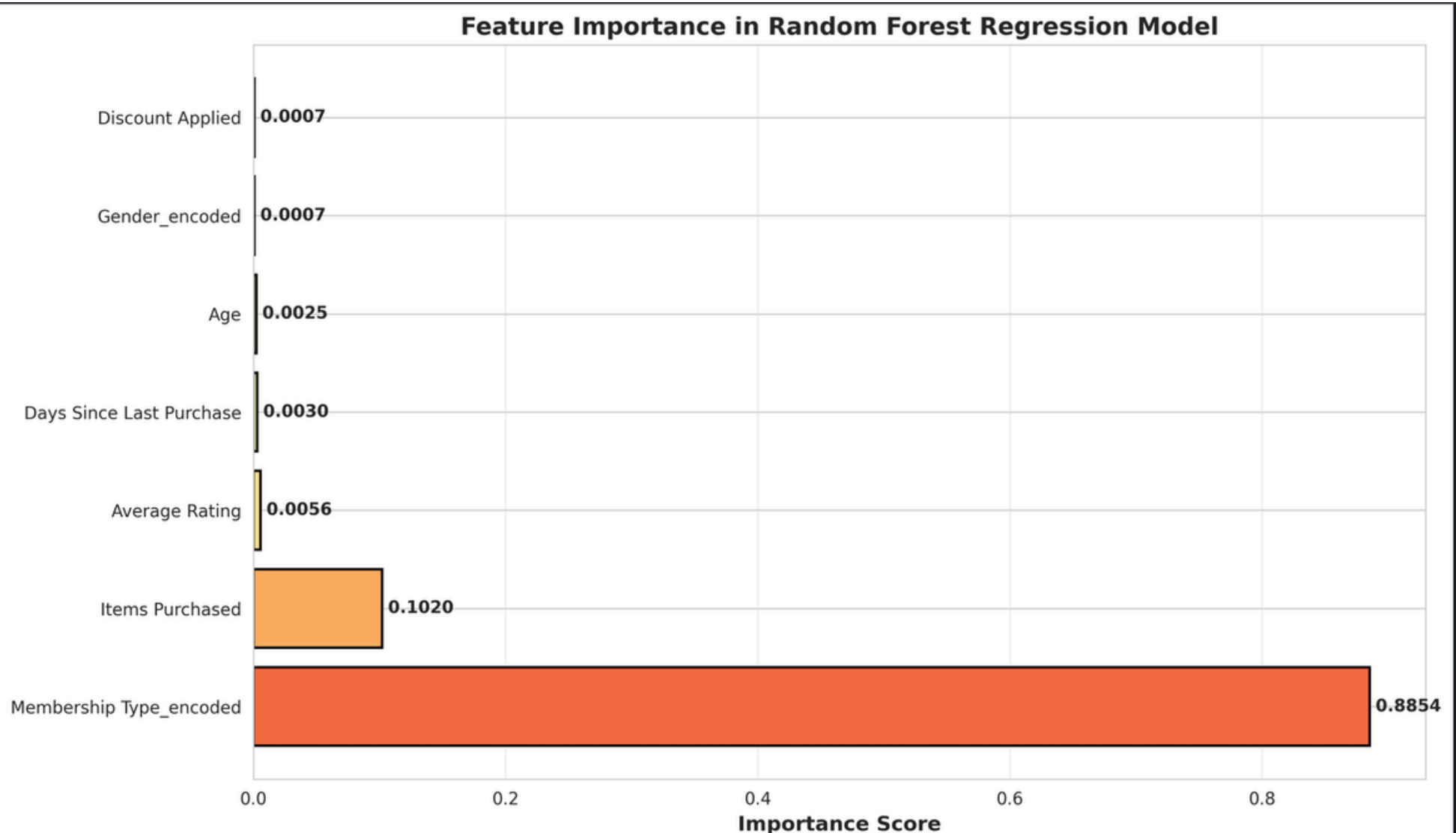
The tests has confirmed key assumptions were reasonably satisfied

- **Overall model is statistically reliable and supports effective data-driven decision-making**

Regression & Interpretation



Classification Interpretation



Conclusion ◀◀

This project analyzed customer engagement and purchasing behavior using statistical and predictive modeling. Key engagement factors such as cart additions, time spent, pages visited, and prior purchases significantly influence purchase probability. Overall, the findings show how data-driven approaches can transform e-commerce activity into actionable strategies for improving conversions, targeting customers, and guiding business decisions.

